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Networking and Information Technology Research and Development
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Federal AI Action Plan

Response to Request for Information

March 15, 2025

Provided to:

National Science Foundation
Faisal D'Souza
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March 15, 2025

Faisal D'Souza
National Science Foundation
[REDACTED]

Subject: Response to Request for Information for Federal AI Action Plan

Dear Faisal D'Souza:

The United States is sitting at the crossroads of economic and national security and the opportunity to achieve AI dominance. Failure to achieve dominance is to cede a once in a generation opportunity to drive massive private and public sector productivity, efficiency, and innovation. With this imperative in mind, Guidehouse is enthusiastically providing our response to the Federal AI Action Plan Request for Information to the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO). Our response reflects our full commitment to the Trump Administration's bold AI path, and our experienced and informed recommendation for a successful Federal AI Action Plan that achieves AI dominance.

Guidehouse is one of the largest technology and business advisory firms in the world, but also one of the youngest. Founded only six years ago, we have achieved global scale, demonstrated client impact across commercial healthcare, energy, and financial services, as well as public sector clients, and the emergence of a recognizable brand that could never have happened a decade ago. This happened today because we launched the company to be a disruptor leveraging a fully cloud-based, AI-enabled environment. Our more than 18,000 employees collaborate with innovative business leaders to achieve transformational changes that shape our client's success.

We appreciate the opportunity to present our recommendations and ideas for the development of a Federal AI Action Plan. Our points of contact for this response are our contracts lead, Jill Raphael-Terry, at [REDACTED] or me at [REDACTED]

Sincerely,

Scott McIntyre
Guidehouse, Chief Executive Officer

Stuart Brown
Guidehouse, Global Technology Lead

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1.0 Introduction

The Federal Artificial Intelligence (AI) Action Plan under the Trump Administration emerges at a pivotal moment, aiming to cement the United States' dominance in a fiercely competitive global AI landscape. AI is reshaping economies, redefining national security, and revolutionizing public and private sector service delivery. As the United States (U.S.) seeks to accelerate its leadership in AI, the federal government must adopt a forward-looking approach that fosters innovation, enhances national competitiveness, and promotes responsible AI development. With adversaries like China investing heavily in state-driven AI strategies, the U.S. faces an urgent challenge to outpace rivals through innovation, agility, and economic might. President Trump has underscored this plan as a cornerstone for enhancing America's technological supremacy, prioritizing outcomes that fuel growth and securing national interests. Leveraging a deregulatory ethos and public-private dynamism, the AI Action Plan seeks to unleash AI's potential by enabling and scaling innovation and adoption across critical domains.

2.0 Our Approach and Recommendations

America's dominance in AI will require more than identifying AI use cases and investing in the development of AI solutions. Thus, Guidehouse took a holistic approach in recommending policy actions, and we have organized our recommended policy actions into the following broad, cross-cutting categories:

- ***Charting America's AI Future: Creating a Dynamic Policy Ecosystem for Technological Leadership and Public Value*** – building a flexible, outcome-focused AI policy framework that drives U.S. technological leadership while driving public trust, economic growth, and responsible deployment.
- ***Powering America's AI Frontier: Harnessing Data, Security, and Infrastructure*** – producing accessible, high-quality data and powerful infrastructure to unlock America's AI potential while protecting the security of U.S. AI solutions from adversaries while also leveraging AI to enhance U.S. cybersecurity.
- ***Sustaining and Accelerating America's AI Dominance: Mobilizing Resources to Accelerating AI Development and Usage*** – coordinating AI efforts, enhancing AI education, upskilling the workforce, and reforming immigration policies to maintain AI leadership, drive innovation, and prevent talent shortages and economic stratification.

Our recommended policy actions across these categories accelerate America's advancement in AI both in the short-term and make it sustainable in the long-term while staying vigilant on America's competitive place in the world.

2.1 ***Charting America's AI Future: Creating a Dynamic Policy Ecosystem for Technological Leadership and Public Value***

Current Environment:

As the Administration contemplates its strategic approach to AI, the emerging policy landscape presents a critical opportunity to balance innovation, economic competitiveness, and responsible guardrails on security and transparency that protect the U.S.'s interests while accelerating its lead in innovation. There is no comprehensive federal law specifically regulating AI. Instead, public

and private organizations must navigate a complex web of federal mandates, international (e.g., the EU's General Data Protection Regulation) and state-specific data privacy laws (e.g., California Consumer Privacy Act). These regulations, along with rules interpreted as applying to automated decision-making systems and sector-specific compliance requirements¹, can add burdensome costs to AI development and adoption. This often entrenches incumbents who can bear the costs and stifles small business innovation.

The Federal AI Action Plan must articulate a dynamic framework that reduces regulatory friction, addressing the labyrinth of regulatory hurdles, compliance barriers, and risk-averse bureaucratic processes that currently restrict AI's ability to solve real-world problems. Simultaneously, it should elevate the federal government's strategic role in guiding AI development through public-private partnerships and light-touch oversight focused on outcomes that protect the public's interests. This will enable faster translation of AI capabilities into practical applications both across industry and the public sector.

Recommended Action(s):

- **Create structured collaboration mechanisms between agencies and the industries they regulate:** Historically, when policymakers have prioritized enabling innovation over strict enforcement, technological progress has flourished. In the state of Michigan, a grant program was established to support small manufacturers in adopting AI technologies. The initiative has successfully financed over 130 projects, leading to increased productivity and competitiveness in the manufacturing sector². The result was an increase in revenue of \$174 million for \$3 million in grants awarded.

Centralized National Artificial Intelligence Research Institutes or a similar body, in collaboration with federal agencies, should lead these partnerships federally, leveraging sector-specific expertise to drive targeted support. Direct funding or incentives and access to computational resources and/or infrastructure support for small businesses will lower barriers to AI implementation in industries where it can provide the highest economic and societal value. Agencies will have a clear mandate of enabling innovation while protecting the public's interest with performance, economic, and workforce impact metrics to evaluate AI use cases in industries within their areas of expertise.

- Establish formal AI sandboxes between public and private commercial businesses, colleges and universities and relevant government agencies: Not all organizations can make the investment needed to fund AI initiatives, have the requisite internal expertise, or have a strategic understanding of how to deploy AI. For this purpose, agencies can create AI regulatory sandboxes—controlled testing environments that enable rapid innovation under smart oversight. This approach shifts the regulatory paradigm from a traditionally restrictive

¹ https://www.ftc.gov/system/files/ftc_gov/pdf/EEOC-CRT-FTC-CFPB-AI-Joint-Statement%28final%29.pdf

² <https://www.michiganbusiness.org/49d4f8/globalassets/documents/partners-toolbox-training/industry-4.0-presentation-6-13-2023.pdf>

pre-approval model to one that prioritizes innovation by permitting activities by default unless they are specifically prohibited.

For example, PrecisionFDA, a “secure, collaborative, high-performance compute environment” with a sandbox to fine-tune generative AI models, allows for public participation in the advancement of the use of AI through hackathons, where research performed in the sandbox helps establish technological feasibility for all participants, lowering the cost of development³. Through centralized sandboxes for each domain or industry (e.g., autonomous aviation systems, healthcare, defense technologies), a standardized reference for training, tuning, and testing available across use cases to all participants helps rapidly advance the iterative research and development (R&D) process, breaking silos where progress may have been limited by barriers the federal government is uniquely equipped to break.

Additionally, the government should partner with industry leaders, tapping into their expertise to co-design sandboxes best suited for ideation and experimentation, bridging the gap between research and practical applications to ensure pilots reflect real-world use cases rather than overly academic exercises. Where regulatory arbitrage⁴ is a risk, the U.S. should cooperate with international partners for sandboxes across borders to facilitate international collaboration and harmonize regulations across jurisdictions. As one example, the EU Metis Sandbox has successfully allowed data holders to test datasets for interoperability⁵.

Downstream for sandboxes, agencies should establish regulatory test zones for successful pilots where companies can deploy AI solutions under temporary waivers before full regulatory approval in heavily regulated industries such as aviation, defense, or healthcare.

- **Streamline AI procurement and deployment for public services:** The benefits of sandboxes flows both ways. By collaborating with private-sector innovators to test AI systems under real-world conditions or scenarios, agencies can leverage these insights for public sector AI use cases – accelerating service efficiency, fraud detection, and automated decision-making, among others. As AI technologies are pre-tested, the government should modernize procurement rules to allow for faster approval of these AI-driven solutions, allowing agencies to rapidly integrate proven AI tools from the private sector.

The federal government is one of the largest purchasers of AI systems, spending an estimated \$3.3 billion on AI-related contracts in 2022⁶. Even so, the Federal Acquisition Regulation (FAR) and agency-specific procurement processes can take years to finalize AI contracts, hindering agencies from keeping pace with innovation. Additionally, current procurement policies often require vendors to have extensive past performance records, making it difficult for startups and emerging AI service firms to compete. To accelerate AI acquisition pipelines,

³ <https://www.fda.gov/media/182815/download>

⁴ https://scholarship.law.upenn.edu/cgi/viewcontent.cgi?article=3570&context=faculty_scholarship

⁵ <https://pro.europeana.eu/post/how-aggregators-work-with-the-metis-sandbox-to-deliver-high-quality-data>

⁶ https://ai.gov/wp-content/uploads/2023/11/Recommendations_AIs-Procurement-Challenge.pdf

the federal government should establish pre-vetted AI contract vehicles that allow agencies to fast-track integration of proven AI tools, with a “Fast Track” program that enables agencies to experiment and deploy AI solutions they have already tested within their own or other agencies’ regulatory sandboxes. These AI contract vehicles should include criteria that allow startups and emerging firms to compete with entrenched players. Moreover, AI technology and service firms that contribute to value-driven R&D within government sandboxes or through public-private partnerships should be given priority to further incentivize private-sector participation.

- **Create cross-agency AI innovation networks via “push-pull” mechanisms for AI adoption:** To accelerate AI adoption across the federal government, agencies need a structured “push-pull” mechanism – one that pushes value-driven AI innovations into scalable deployment pipelines while simultaneously pulling agency-specific use cases into centralized R&D initiatives. Creating a hub-and-spoke model effectively allows for centralized AI R&D to efficiently spread across agencies while allowing agencies to pull solutions that match their specific needs. This can be established through a Federal AI Center of Excellence to serve as the primary research and development hub in partnership with national laboratories, academic institutions, and the sandboxes above. The benefits are multi-pronged: agencies would have AI toolkits and frameworks that they can adopt and customize for their specific needs, a shared services model establishes reusable components and/or interoperability standards for common use cases across agencies such as procurement, budgeting, and HR functions; and ensures that AI projects have deployment pathways across multiple agencies, avoiding duplication of R&D efforts.

To enable the greatest utility to the American taxpayer, agencies can submit specific use cases within an allocated budget, promoting responsible value generation while considering both effort and impact. Priority should be given to agencies through pooled funding mechanisms, where multiple agencies with overlapping needs can co-finance AI innovation.

Outcome(s):

The recommended actions represent a comprehensive approach to accelerating AI adoption within agencies and across the industries they regulate, fostering an approach to innovation that balances accelerating value-driven usage of AI and oversight that protects the public’s interests. These recommendations can produce the following outcomes:

- **Structured collaboration between agencies and industries:** This approach is expected to transform the regulatory landscape, encouraging innovation and reducing compliance costs, leading to substantial economic gains while promoting safety and efficacy. Experiences from agencies that have already adopted AI technologies demonstrate that significant reductions in processing times and costs are achievable.^{7,8}
- **Controlled environments for AI testing:** Establishing these environments will accelerate the time-to-market for AI applications by allowing stakeholders to rigorously test and adopt innovations faster.

⁷ [*Agency Boosts Efficiency with Automation and AWS Cloud | Guidehouse*](#)

⁸ [*Machine Learning to Predict Government Fraud | Guidehouse*](#)

- **Modernized procurement processes:** Streamlining these processes will enable rapid integration of AI solutions into public sector operations, cutting costs and inefficiencies, and improving service delivery and outcomes for citizens.
- **Cross-agency collaboration and knowledge-sharing:** Leveraging successful AI implementations across agencies will reduce duplication of effort, speed up the adoption of proven solutions, and create a powerful ecosystem, strengthening America's competitive position in the global AI landscape.

2.2 Powering America's AI Frontier: Harnessing Data, Security, and Infrastructure

Current Environment:

The expansion of AI has led to the drastic and rapid demand for computing power. AI systems are constrained by two critical factors: the availability of massive amounts of computational resources and access to high-quality data. The technology's effectiveness scales exponentially with data volume and computational resources, a reality that makes securing these fundamental elements crucial to maintaining the U.S.' lead in AI innovation.

The economics are stark: the cost of compute remains the primary barrier to AI advancement, where training a single large language model can easily approach millions of dollars. The AI Action Plan should prioritize scaling the nation's energy infrastructure, such as the announcement of the Stargate Project⁹, and driving down the cost of compute through investments in data centers, semiconductor production, quantum computing, and chip innovation.

The U.S. must also break down barriers to datasets across government agencies. Despite initiatives like the Federal Data Strategy and the OPEN Government Data Act, critical datasets remain siloed by inconsistent formats, inadequate documentation, and lack of strategic planning of usability and dissemination. AI systems cannot reach their full potential if they are starved of the high-quality data necessary for training and evaluation.

AI dominance is pivotal to military strategy, economic productivity, and scientific discovery. Competitor nations are making unprecedented investments in AI infrastructure. China is said to have "accounted for more than half of all global investment over the last five years"¹⁰ to become the global AI leader by 2030, focusing heavily on chip production and supercomputing facilities. Without urgent and large-scale investments in AI infrastructure and deliberate policy action to address security risks, the U.S. risks falling behind nations that are aggressively scaling their AI capabilities, both in research and deployment.

History has shown that when the stakes are highest, the U.S. succeeds by mobilizing its talent, industry, and government in a unified effort. During World War II, the Manhattan Project was not just about developing atomic weapons—it was about securing the future. The Apollo Program ensured not only that the U.S. won the space race, but that it set the foundation for

⁹ [*The Stargate Project: A Manhattan Project for the AI era*](#)

¹⁰ [*Made in China 2025: Xi Jinping's plan to turn China into the AI world leader - ABC News*](#)

decades of aerospace and computational advancements. AI today is no different. It demands a nationally coordinated Action Plan on the scale of these historic efforts.

Recommended Action(s):

- **Make High-Quality AI-Ready Data Available:** The availability of high-quality data is fueling AI advancements. The AI Action Plan should empower the National AI Research Resource (NAIRR) or a similar body to not just make data available, but to prioritize data quality standards required for cutting-edge AI development free from bias or misinformation. Several government agencies already have initiatives that aim to release their data; however, these efforts are fractured, without consistency in access or usability. Even if an agency wishes to share its data, the presence of numerous regulations, unclear guidance, and occasionally conflicting directives may cause the agency to refrain from taking any risks. Datasets should have the critical context, structure, and reliability necessary for AI model training and development. NAIRR, in partnership with similar governing bodies such as NIST and OSTP, should take the lead in national AI data standards with sharable datasets across agencies, with tiered levels of access (depending on security level of data) both intra-agency and to the public. This includes creating policy guidelines, procedures, and developing data-sharing standards for government agencies to make AI-ready data accessible. A dynamic policy framework is necessary to be responsive to AI advancements, preventing regulatory misalignment. The U.S., as part of its AI Action Plan, should invest in making AI-ready data a national strategic asset.
- **Address Current Energy Bottlenecks and Proactively Prepare for Future Energy Challenges:** AI compute is energy intensive, with the computational power needed to power next-generation AI systems estimated to be measured in gigawatts¹¹. Securing stable, cost-effective power sources, such as nuclear, liquefied natural gas, renewables, and advanced grid technologies, is crucial for long-term AI leadership. Not only does this keep the U.S. at the forefront of innovation, but it also powers the economy, with the Department of Energy estimating that the U.S. will require 19,000 new electrical and electronics engineers annually, and an 11-percent annual increase in electricians¹².

The Administration should direct the Department of Energy to modernize its current framework surrounding energy infrastructure, with an AI-specific roadmap. Opportunities exist for expedited licensing of AI-focused infrastructure, such as a specialized fast-track regulatory pathway at the Nuclear Regulatory Commission for data center and AI computing projects, production tax credits for AI-focused expansion that mirrors those used to stimulate wind and solar growth, and federal grants programs supporting research into energy storage technologies that can support AI's variable energy demands.

The current U.S. energy grid was not designed to accommodate the unprecedented power demands of AI computing. Upgrading the grid with AI-optimized, high-throughput energy systems is critical for furthering innovation by American innovators. The Action Plan should include investments in modernized grid networks, including investments in AI-driven grid

¹¹ https://www.rand.org/pubs/research_reports/RR43572-1.html

¹² https://www.energy.gov/sites/default/files/2024-10/USEER%202024_COMPLETE_1002.pdf

optimization. By creating a comprehensive plan to rapidly enhance and modernize the U.S. energy grid, the nation can enable equitable access for all companies, fostering a level playing field in AI advancements. This initiative will not only support the growth of the AI sector but also contribute to the overall resilience and efficiency of the national energy infrastructure.

- **Establish AI Resilience as a National Security Priority:** As AI becomes a core pillar of economic and defense competitiveness, energy resilience is a critical security concern. AI compute infrastructure must be protected from energy shortages, cyber threats, and adversarial interference. The Action Plan should address the independence of the AI-energy supply chain from adversarial energy providers.

Similarly, The Defense Advanced Research Projects Agency (DARPA), National Science Foundation and Intelligence Advanced Research Projects Activity should dramatically scale investments in AI Assurance and counter-adversarial techniques, with a focus on preemptive threat detection and neutralization. AI systems are already capable of outperforming humans in certain tasks, with super-intelligent systems expected soon after. Strategic funding in AI security can preemptively mitigate the risk of large-scale societal disruptions or turbulence. The Administration should build a distributed network of laboratories capable of rapid response testing and decommissioning of potentially dangerous AI technologies. Internationally, the U.S. should lead with binding standards for identifying and neutralizing AI models with potential weaponization risks, with real-time monitoring systems to track evasion.

National Security agencies should develop the capacity to test and evaluate models at a more granular level, with tiers or classifications depending on their potential risk that would trigger more stringent national security reviews. For those classified as closest to “Artificial General Intelligence” (AGI), the U.S. needs to monitor their development and prohibit the development of fully autonomous AI systems capable of self-modification without human oversight. Previous successful initiatives, such as DARPA’s Media Forensics program, have demonstrated the effectiveness of proactive investments in combating adversarial uses of AI, such as deepfake technologies. Additionally, incentives like the Center for Human-Compatible AI at UC Berkley highlight the importance of aligning AGI systems with human values and ethical norms, ensuring that advanced AI operates transparently and ethically. By funding comprehensive R&D in these areas, the U.S. can outmaneuver malicious actors, create new defensive measures, and build robust testing frameworks with wide applicability.

- **Establish a U.S. led global AI development and governance framework:** As AI technologies transcend borders, nations are competing for leadership through varying approaches to regulation, national security, and intellectual property policies. To effectively promote American values and those of like-minded nations, international cooperation and harmonization are essential, ensuring that democratic values and economic freedoms remain central to AI’s global trajectory.

The policy framework must be adaptive, recognizing that AI technologies will continue to evolve in ways that are hard to fully anticipate. Traditional regulatory frameworks struggle to keep pace with AI’s rapid evolution, with current international regulations fragmented and potentially restrictive. In their “Economic Blueprint,” OpenAI states that “There’s an estimated \$175 billion sitting in global funds awaiting investment in AI projects, and if the U.S. doesn’t attract those funds, they will flow to China-backed projects —strengthening the

Chinese Communist Party's global influence"¹³. By collaborating with allies in the Group of Seven, Organisation for Economic Co-operation and Development, and Quad nations that share similar values and goals regarding AI governance, the U.S. can create a unified front that promotes transparent, secure, and economically beneficial strategic AI frameworks (e.g., a tiered investment tax incentive structure, market access benefits, AI talent development or recruitment) such that entrepreneurs and technologists fund innovation and development of frontier models in the U.S. and not with adversaries.

As of August 2024, China already has over 180 Large Language Models (LLMs) approved for general use in the public sector¹⁴. To prevent the domination of Chinese LLMs in the market, this global framework should have a goal of promoting the export of U.S. aligned- and U.S.-built frontier models with allies to promote their use, helping U.S. allies' local economies through U.S.-aligned AI technology.

The U.S. should lead in developing AI development, governance, and safety standards as guidelines that protect fundamental rights while maintaining the rapid innovation cycle essential to technological leadership. This coordinated strategy would prevent individual states or foreign jurisdictions from creating a maze of conflicting requirements from fragmenting the AI market and create unnecessary barriers to entry for U.S. companies.

- **Strengthen export controls and technology transfer restrictions to reduce strategic dependency on adversarial nations with potentially malicious motives:** Robust export controls on AI-enabling technology are necessary to prevent intellectual property leakage and slow development from adversarial nations and state actors. Claude, one of the current leaders in frontier models globally, shared that their most recent model demonstrated the concerning ability to support aspects of biological weapons development¹⁵. In the wrong hands, AI can develop biological and nuclear weapons that pose a national security threat to the U.S. and the globe. The U.S. should implement rigorous controls on the export of AI models and model weights, training infrastructures, and specialized hardware. Governing access to computational resources is crucial for advanced AI development is another critical element of this framework. Implementing strict licensing and access controls for high-performance computing will enable authorities to monitor and regulate resource access.

Outcome(s):

The recommended policy actions are aimed at enhancing the availability of high-quality data, addressing current and future energy needs, and establishing a safe and secure AI infrastructure, positioning the U.S. as a global leader in AI innovation and security. Implementing these recommendations will yield several outcomes:

- **Enhanced data quality and accessibility:** NAIRR, in collaboration with NIST and OSTP, will create and maintain dynamic policy guidelines and data-sharing standards, enabling

¹³ <https://cdn.openai.com/global-affairs/openai-us-economicblueprint-feb-2025-edu-update.pdf>

¹⁴ <https://technode.com/2024/08/13/china-now-has-over-180-llms-approved-for-general-use/>

¹⁵ <https://assets.anthropic.com/m/785e231869ea8b3b/original/claude-3-7-sonnet-system-card.pdf>

structured, reliable, and readily available datasets for AI development. Standardization will eliminate the current fragmentation and inconsistency in data access and usability, providing researchers, startups, and established companies with the necessary resources to develop innovative AI models.

- **Stable and cost-effective energy infrastructure:** Recognizing that AI compute is energy-intensive, the Administration will modernize the current energy framework, creating an AI-specific roadmap. By upgrading the grid with AI-optimized, high-throughput energy systems, the U.S. will support widespread AI innovation and provide all companies, large and small, equitable access to the necessary computational resources.
- **National Security and Resilience:** Establishing AI resilience as a national security priority will safeguard AI compute infrastructure from energy shortages, cyber threats, and adversarial interference. Building monitoring capabilities for frontier AI models mitigates national security risks before they materialize into actual threats. By significantly increasing funding for AI Assurance and counter-adversarial AI techniques, the U.S. will develop robust defense mechanisms to safeguard AI energy access. Joint AI-energy security task forces will be established to protect critical AI infrastructure from both cyber and physical threats, ensuring the nation's AI capabilities are secure and reliable.
- **U.S.-led Global AI Leadership:** A U.S.-led global AI development and governance approach will establish democratic nations as the primary architects of international AI standards. By convening like-minded partners and creating a compelling alternative to technological models from nations seeking to rival U.S. dominance, the U.S. can effectively prevent those nations from capturing the regulatory narrative. U.S. leadership will ensure that AI development remains aligned with principles of transparency, individual rights, and economic freedom.
- **Strategic Resource Alignment:** Reducing dependency internationally for AI technology from potentially hostile actors preserves the U.S.'s technological edge while preventing the proliferation of AI technologies that could compromise national security. This strategic approach to technology protection will strengthen the U.S.' diplomatic position, enabling the administration to build deeper technological partnerships with allied nations while maintaining essential safeguards against potential threats.

2.3 Sustaining and Accelerating America's AI Dominance: Mobilizing Resources to Accelerating AI Development and Usage

Current Environment:

The release of ChatGPT-3.5 in late 2022 marked a turning point in technological history, demonstrating AI's unprecedented ability to generate human-like content across text, video, audio, images, and code. This breakthrough, followed by rapid advancements in generative AI, has fundamentally shifted the conversation about the future of work and workforce development.

Sam Altman predicts that 2025 will be the year AI agents join the workforce¹⁶. The versatility of new AI capabilities presents both opportunities and challenges. While they demonstrate remarkable capabilities in automating certain tasks, they create new possibilities for augmenting human productivity and creating entirely new categories of work. The range of AI possibilities is

¹⁶ <https://blog.samaltman.com/reflections>

exceedingly broad, requiring a proactive policy framework that harness AI's potential while empowering the American workforce.

U.S. leadership in AI requires more than technological supremacy. It requires a national strategy to cultivate AI talent and create efficient channels for innovation and resource allocation between public and private sectors. Without these mechanisms, the nation risks a shortage of skilled AI professionals that could stall innovation and weaken the nation's competitive edge, falling behind global competitors.

Recommended Action(s):

- **Launch an AI Production Board:** To realize the objectives of the AI Action Plan as outlined in Executive Order 14179, the U.S. requires a coordinating entity akin to the War Production Board of World War II, reimagined as an AI Production Board (AIPB). This organization would serve as a public-private nexus, galvanizing industry, academia, and government to accelerate AI development and deployment in a manner consistent with the administration's deregulatory vision. The AIPB would not impose mandates but rather act as a catalyst, channeling resources, aligning priorities, and incentivizing breakthroughs to outpace global competitors like China, where state-driven strategies threaten U.S. technological supremacy.

The AIPB's necessity stems from the fragmented nature of current AI efforts, which risk diluting America's competitive edge absent strategic orchestration. By offering tax incentives, facilitating access to federal computing resources, and establishing voluntary benchmarks – such as leading global AI patent filings by 20% by 2027 or deploying AI to save \$50 billion in federal costs by 2030 – the AIPB could drive market-led innovation without stifling entrepreneurial freedom. Drawing from historical precedent, the War Production Board transformed disparate industries into a unified war machine; similarly, an AIPB could harness America's tech giants (e.g., NVIDIA, Microsoft), startups, and research institutions to prioritize AI applications critical to economic competitiveness and national security, such as advanced manufacturing or cybersecurity.

Moreover, the AIPB would address workforce and resource gaps by partnering with private entities to train millions in AI skills and pool anonymized federal datasets for developers, all while avoiding the bureaucratic overreach the administration seeks to eliminate. This lean, industry-empowered model positions the U.S. as the "Arsenal of AI," securing human flourishing and economic leadership as envisioned in the Executive Order, without sacrificing the agility of a free-market approach.

- **Accelerate AI Education and Training at Scale:** The development of a National AI Education Initiative faces significant hurdles, including inadequate funding, a shortage of qualified educators, and unequal access to resources across educational levels. Integrating AI into K-12 STEM curricula demands substantial investment to train teachers and update materials, yet budget constraints often limit progress.

To address these challenges, integrating AI into K-12 STEM requires federal funding to equip schools, train educators, and modernize curricula. For higher education and vocational training, scaling AI degree pathways need targeted investment such as funding for AI research centers at universities and grants for AI students. Simultaneously, the government should provide federal funding to increase the number of AI professors to tackle the faculty shortage

to meet demands for AI skills. These concrete steps aim to build a robust AI education ecosystem.

- **Upskill the Existing Workforce for AI:** Implementing a national AI education and workforce transition initiative faces significant challenges, including funding shortages, a lack of trained educators, and the rapid pace of AI-driven job displacement. The International Monetary Fund estimates that almost 40% of jobs will be impacted by AI, with up to 60% in advanced economies like the U.S.¹⁷, risking mass unemployment in sectors like manufacturing without intervention. A 2020 House Budget Committee report warned, “AI-enabled automation could contribute to further wage inequality”¹⁸ if workers aren’t reskilled. To counter displacement, the Federal government should provide tax incentives for corporations who develop AI reskilling programs in industries where many jobs could change due to automation and agentic AI capabilities.

A study by Georgetown’s Center for Security and Emerging Technology indicates that the rapid growth in computer science enrollments has not been matched by a similar increase in the number of computer science instructors¹⁹. This imbalance may pose a challenge to expanding the pool of AI developers. Drawing upon South Korea’s AI curriculum success²⁰, fast-track AI certifications could train thousands of workers yearly. In addition, the Federal government should consider incentives to support public-private AI apprenticeship programs providing hands-on AI experience. These actions aim to turn AI’s disruption into opportunity.

- **Reform High-Skilled Immigration Pathways for AI Talent:** Retaining top AI talent in the U.S. faces challenges of global competition and slow regulatory processes. A 2023 National Foundation for American Policy report notes that 70% of graduate students in fields related to AI at U.S. colleges and universities are international²¹, yet many leave due to visa backlogs. The U.S. risks losing its competitive edge in AI to other nations as countries like Canada offer faster pathways for AI-focused graduate students currently at U.S. colleges and universities to be able to work after completing their studies.

To address this, the U.S. should expand an AI-specific visa program, modeled on the U.K.’s system but designed for U.S. national security needs. This could fast-track the recruitment of AI experts into the country’s resource pool each year. Targeting professionals currently enrolled in U.S. colleges and universities with AI patents or advanced degrees, this visa would quickly bolster the U.S. workforce. In addition, the federal government should streamline the visa processes for AI experts to address a sizable case backlog recently reported by the

¹⁷ <https://www.imf.org/en/Blogs/Articles/2024/01/14/ai-will-transform-the-global-economy-lets-make-sure-it-benefits-humanity>

¹⁸ [Machines, Artificial Intelligence, & the Workforce: Recovering & Readyng Our Economy for the Future | House Budget Committee Democrats](#)

¹⁹ [AI Faculty Shortages | Center for Security and Emerging Technology](#)

²⁰ [A Guide To South Korea's Revolutionary AI Education: Shaping Tomorrow's Tech Leaders](#)

²¹ [AI-AND-IMMIGRANTS.NFAP-Policy-Brief.2023.pdf](#)

USCIS²². This will make it easier for top AI graduate students and researchers to gain permanent residency in the U.S. after completing their education.

Outcome(s):

AI leadership will be determined by who has the talent and capacity to execute at scale. The recommended policy actions are aimed at building and utilizing a highly skilled workforce that can invent, govern, and apply AI technologies across the entire economy. By incentivizing breakthroughs through public-private partnerships, investing in education, and attracting the top AI talent globally, the Administration can allow both workers and the nation thrive in an era of AI transformation. Implementing these recommendations will yield several outcomes:

- **Rapid Channeling of Resources:** Coordinating industry, academia, and government efforts, offering tax incentives, and facilitating access to resources through a new AI Production Board drives innovation and economic leadership while avoiding excessive regulation.
- **Stronger Pipeline of Domestic AI Talent:** Expanding AI education develops the next generation of workers who are equipped to weave AI into everyday life and compete at the highest level.
- **Reduced AI Talent Shortages Across Industries:** By upskilling the existing workforce, filling AI job openings will become easier and drive widespread AI-powered innovation in sectors beyond just tech.
- **Increased AI Workforce Participation Across Economic Demographics:** Structured AI apprenticeships and technical training programs provide equitable workforce access to AI capabilities, preventing AI-driven economic stratification.
- **A Stronger Retention of Foreign AI Talent:** Select AI-specific immigration reforms provide an additional source for top AI talent produced by U.S. colleges and universities to stay within the country, rather than fueling foreign AI advancements.

3.0 About Guidehouse

Guidehouse is an innovative, private equity backed start up that has quickly disrupted our industry with different business, incentive, and solutioning models. Guidehouse is an AI integrator, AI developer, AI trainer, AI investor, and importantly an aggressive AI user. Based in the U.S., Guidehouse employs more than 18,000 professionals, and delivers end-to-end AI solutions—spanning data management, intelligent automation, advanced analytics, digital modernization, and infrastructure optimization—that empower commercial companies and federal agencies to lead in a deregulated, market-driven AI landscape.

Guidehouse is free from the obvious conflicts associated with an industrial base of longstanding federal contractors and consultants who built and sustain the legacy technology architecture of the federal government that is so expensive and outdated given commercial alternatives. Having guided Fortune 500 clients through complex transformations, Guidehouse offers the strategic

²² According to a 2023 report, the USCIS has a net backlog of 4.3 million: [Completing an Unprecedented 10 Million Immigration Cases in Fiscal Year 2023, USCIS Reduced Its Backlog for the First Time in Over a Decade | USCIS](#)

insight and expertise to align federal priorities—economic competitiveness, national security, and workforce readiness—with private-sector dynamism.

We want to leverage our commercial and public sector AI capabilities and investment to collaborate with the U.S. Government in crafting and executing the Federal AI Action Plan under President Trump’s Executive Order 14179, bringing additional relevant experience in government policy, economic strategy and impact analysis, energy transformation, and cutting-edge technology innovation. We endeavor to support the Administration’s vision of American AI dominance, free from bureaucratic overreach, to accelerate innovation and counter foreign state-led advances. We are fully committed to the Trump Administration’s bold AI path, excited to drive the federal government’s journey toward a future where U.S. leadership in AI enables human flourishing and global preeminence.

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